Simultaneous Determination of the Branching Fractions of $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ in LHCb Run 3 - Final Report

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This dissertation presents a simultaneous determination of the CP-averaged branching fractions of $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ using LHCb Run 3 data. These colour-favoured $b \to c\bar{u}q$ transitions provide a sensitive probe of QCD factorisation predictions, which currently exhibit tension with experimental measurements. The fully software-based Run 3 trigger enables efficient selection of hadronic final states and allows normalisation to the clean decay $B^+ \to J/\psi K^+$. The measured branching fractions obtained in this analysis are $\mathcal{B}(\bar{B}^0 \to D^+\pi^-) = (2.165 \pm 0.007_{\text{stat}} \pm 0.032_{\text{exp}} \pm 0.069_{\text{ext}}) \times 10^{-3}$ and $\mathcal{B}(\bar{B}^0 \to D^+K^-) = (1.933 \pm 0.014_{\text{stat}} \pm 0.024_{\text{exp}} \pm 0.062_{\text{ext}}) \times 10^{-4}$. Projections to the full Run 3 dataset indicate a statistical precision improvement approaching an order of magnitude compared to the most precise existing single measurements.

I. Introduction

A. The $b \to c\bar{u}q$ Anomaly

Precision measurements of non-leptonic b-meson decays provide direct tests of the Standard Model description of the weak interactions of heavy-quarks. The decays $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ proceed via the colour-favoured tree-level transition $b \to c\bar{u}q$ ($q = d, s$). Their amplitudes are dominated by a single CKM structure, while penguin and annihilation contributions are strongly suppressed [1].

In the heavy-quark limit, QCD factorisation (QCDF) allows short-distance weak dynamics to be separated from long-distance hadronic effects [2]. For $\bar{B} \to D^+\pi^-$ and $\bar{B} \to D^+K^-$ decays, non-factorisable corrections are expected to be small, leading to branching-fraction predictions with limited theoretical uncertainty [3]. Therefore, these modes serve as benchmark channels for testing the factorisation descriptions of heavy-meson decays.

However, comparisons between

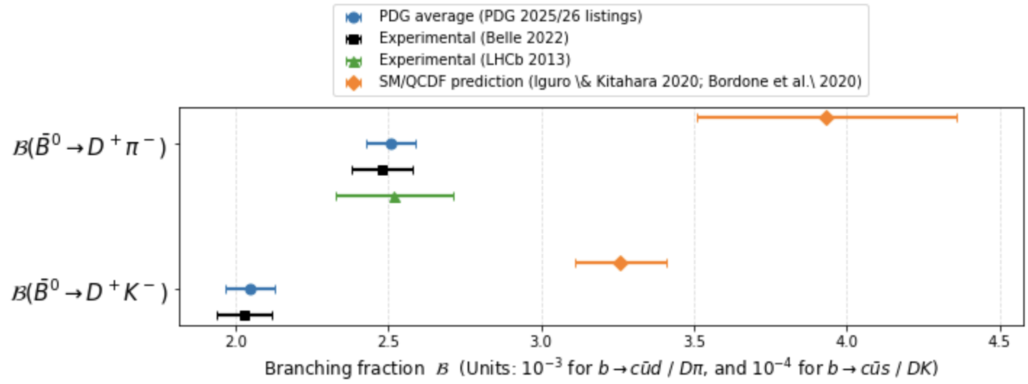

Figure 1: Comparison of experimental measurements and SM/QCDF expectations for $\mathcal{B}(\bar{B}^0 \to D^+\pi^-)$ and $\mathcal{B}(\bar{B}^0 \to D^+K^-)$. The experimental results (PDG averages, Belle, and LHCb) are consistently below the SM/QCDF predictions from Refs. [4, 5], illustrating the long-standing tension in $b \to c\bar{u}q$ transitions.

SM/QCDF predictions and experimental measurements reveal significant tensions, as illustrated in Fig. 1. Using the QCDF calculations of Refs. [4, 5] together with the experimental world averages reported by the PDG, determined by measurements from Belle and LHCb during Runs 1 and 2, the branching fraction $\bar{B}^0 \to D^+K^-$ is found to be lower than the prediction by approximately 5–6σ, while $\bar{B}^0 \to D^+\pi^-$ shows a smaller but still notable discrepancy of about 2–3σ. These measurements are consistent between experiments and lie below the SM/QCDF expectations shown in Fig. 1. This discrepancy is referred to as the $b \to c\bar{u}q$ anomaly. The origin of this discrepancy remains unclear and may reflect limitations in theoretical inputs, experimental systematics, or potentially contributions from physics beyond the Standard Model.

Several previous determinations of $\mathcal{B}(\bar{B}^0 \to D^+h^-)$ were performed through measurements of the ratio $\mathcal{B}(\bar{B}^0 \to D^+K^-)/\mathcal{B}(\bar{B}^0 \to D^+\pi^-)$, rather than as two fully independent branching fractions [6]. In such cases, one hadronic decay is used to normalise the other. Because both channels originate from the same $b \to c\bar{u}q$ transition, any discrepancy affecting the normalisation channel propagates to the final branching fraction, complicating comparisons with QCDF predictions.

A determination that measures the two branching fractions separately and avoids hadronic normalisation provides a more direct comparison with theoretical predictions. The present measurement therefore offers an independent Run 3 determination obtained with the upgraded detector and trigger environment. While not expected to resolve the tension alone, it improves the experimental precision of this comparison.

B. Run 3 Improvements

During LHC Runs 1 and 2, LHCb employed a hardware Level-0 trigger followed by a software high-level trigger [7]. While efficient for leptonic channels, the hardware stage was less effective for hadronic decays. Its decision relied on large transverse energy signals in the calorimeters, requiring high E_T thresholds to reduce the event rate. Many hadronic B decays produce particles with smaller transverse momenta and therefore failed to satisfy these thresholds, leading to reduced efficiency and kinematic biases in hadronic selections.

The Run 3 upgrade replaced the hardware stage with a fully software-based trigger operating at the full LHC collision rate [8]. Real-time tracking and vertex reconstruction are now available at the earliest stage of event selection, allowing trigger decisions to exploit the displaced-vertex signatures of heavy-flavour decays. This significantly improves the efficiency for hadronic B decays, with an increase of roughly a factor of two compared to Run 2 conditions [9].

An important consequence of the Run 3 trigger strategy is the possibility to normalise the measurement to the decay $B^+ \to J/\psi(\mu^+\mu^-)K^+$. This channel provides a clean experimental signature through the dimuon decay of the J/ψ, allowing the decay to be reconstructed with high efficiency. Because the branching fraction of $B^+ \to J/\psi K^+$ is also precisely known, it serves as a well-controlled normalisation mode.

The fully software-based trigger allows both the hadronic signal channels $\bar{B}^0 \to D^+h^-$ and the muonic normalisation decay $B^+ \to J/\psi K^+$ to be selected within a common trigger and reconstruction framework. As a result, the $\bar{B}^0 \to D^+h^-$ branching fractions can be measured individually and can be normalised to $B^+ \to J/\psi K^+$ while avoiding normalisation modes with the same anomaly. This reduces the dependence on less precisely known hadronic inputs and leads to smaller external uncertainties, providing a more robust comparison with theoretical predictions.

C. Aims of this Dissertation

The aim of this dissertation is to determine the flavour-integrated branching fractions

$$\mathcal{B}(\bar{B}^0 \to D^+\pi^-) \quad \text{and} \quad \mathcal{B}(\bar{B}^0 \to D^+K^-)$$

using LHCb Run 3 data. Unless stated otherwise, charge-conjugate processes are implied throughout.

While these branching fractions have been measured previously, this work provides the first determination in the Run 3 dataset using $B^+ \to J/\psi K^+$ as the normalisation channel. The uniform fully software-based trigger environment of Run 3 enables this strategy and provides an internally consis-

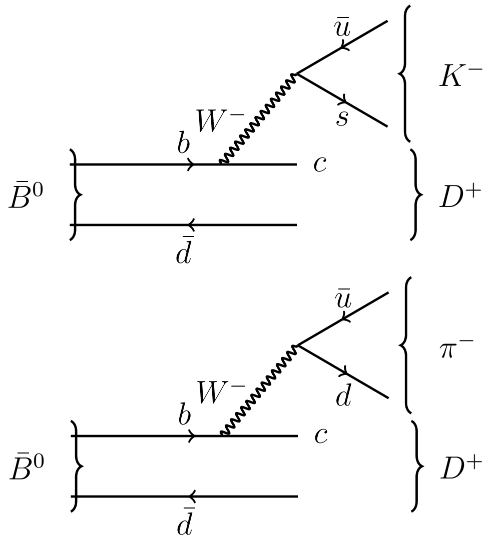

Figure 2: Tree-level diagrams for $\bar{B}^0 \to D^+K^-$ (top) and $\bar{B}^0 \to D^+\pi^-$ (bottom). The exchanged boson is a virtual W^-.

tent framework for testing the $b \to c\bar{u}q$ anomaly.

II. Theoretical Framework

A. Quark-Level Structure

The decays $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ arise from charged-current weak interactions mediated by a virtual W^- boson. The leading-order quark-level process is $b \to cW^-$, followed by the hadronisation of the W^- into a light quark pair $\bar{u}q$, where $q = d$ for the pion mode and $q = s$ for the kaon mode. The spectator $\bar{d}$ quark from the initial $\bar{B}^0$ combines with the charm quark to form the D^+ meson. The corresponding tree-level diagrams are shown in Fig. 2.

The weak amplitude carries the CKM factor $V_{cb}V_{uq}^*$. The relative size of the two decay channels is therefore governed by the ratio $|V_{us}|/|V_{ud}|$, which is well measured [10]. Because $|V_{us}|$ is significantly smaller than $|V_{ud}|$, the D^+K^- final state is Cabibbo suppressed relative to $D^+\pi^-$. This suppression explains the approximate order-of-magnitude difference between the corresponding branching fractions.

Other decay topologies are theoretically possible but strongly suppressed. Loop-induced penguin contributions involve additional CKM suppression and colour factors, while annihilation diagrams require the b and spectator antiquarks to annihilate which is suppressed both dynamically and helicity-wise [11, 12]. Consequently, the decay amplitude is dominated by a single tree diagram, making these channels simpler than many other non-leptonic B decays.

B. QCD Factorisation

Although the weak transition responsible for $\bar{B}^0 \to D^+h^-$ occurs at short distances, the observed final state consists of hadrons formed through non-perturbative QCD dynamics. A central theoretical question is whether these short-distance weak effects can be separated from long-distance strong-interaction contributions.

In the limit where the b quark mass is large when compared to Λ_{QCD}, the characteristic energy scale of non-perturbative strong interactions ($\sim$ 200–300 MeV), QCD factorisation provides a framework in which such a separation becomes possible [3].

Short-distance contributions are calculated perturbatively, whereas long-distance effects are described by hadronic matrix elements.

For colour-favoured tree decays such as $\bar{B}^0 \to D^+h^-$, this separation is expected to work particularly well because the light meson is emitted directly from the virtual W^- boson. At leading order, the decay amplitude can therefore be expressed in terms of a $\bar{B}^0 \to D^+$ transition form factor multiplied by the decay constant of the light meson. Corrections arise from hard-gluon exchange and from terms suppressed by $\Lambda_{\mathrm{QCD}}/m_b$. While these effects cannot be computed with arbitrary precision, they are expected to remain numerically small [3].

The dominant theoretical uncertainties are therefore associated primarily with form-factor inputs and higher-order perturbative corrections rather than with non-factorisable hadronic effects. The theoretical cleanliness of $\bar{B}^0 \to D^+h^-$ decays makes them a sensitive probe of non-leptonic dynamics in the Standard Model.

C. Branching and Fragmentation Fractions

In the absence of flavour tagging, the experimentally accessible observable corresponds to the flavour-integrated branching fraction defined in Eq. (1).

$$\mathcal{B}(B \to f) = \frac{\Gamma(B \to f) + \Gamma(\bar{B} \to \bar{f})}{2\,\Gamma_{\mathrm{tot}}} \quad (1)$$

Although flavour tagging techniques exist at LHCb, they are not used in this analysis as they would reduce the effective statistical power of the sample.

Here $\Gamma(B \to f)$ and $\Gamma(\bar{B} \to \bar{f})$ denote the partial decay widths to CP-conjugate final states, and Γ_{tot} is the total decay width. For the colour-favoured tree decays considered in this work, direct CP asymmetries are expected to be negligible [1]. Neutral B mesons may also undergo B^0–$\bar{B}^0$ mixing before decay but without flavour tagging the measured quantity corresponds to the CP-averaged branching fraction defined above and is unaffected by the mixing.

At a hadron collider, the number of reconstructed decays depends not only on the branching fractions but also on the b-hadron production rate and the integrated luminosity of the dataset. When branching fractions are determined relative to a normalisation channel, these factors largely cancel. Fragmentation fractions, denoted by f_q, describe the probability that a produced b quark hadronises into a particular B meson [13]. In this analysis, the $\bar{B}^0$ signal modes are normalised to a B^+ decay. The quantities f_d and f_u denote the probabilities that a b quark forms a $\bar{B}^0$ or a B^+ meson, respectively. Under the isospin symmetry of the strong interaction, these probabilities are expected to be approximately equal ($f_d \simeq f_u$). Experimental measurements are consistent with this assumption [10].

III. Experimental Setup

The data analysed in this work were collected at the Large Hadron Collider (LHC), a 27 km circumference proton–proton accelerator located at CERN [14]. During Run 3, the LHC operated at a centre-of-mass energy of $\sqrt{s} = 13.6$ TeV with bunch crossings separated by 25 ns. At these energies, heavy-flavour quarks are produced in large numbers via strong-interaction processes, yielding large samples of $b\bar{b}$ pairs suitable for precision flavour-physics studies.

The kinematic properties of heavy-flavour production at the LHC favour the forward emission of b hadrons. The LHCb experiment is specifically designed to exploit this forward production.

A. The LHCb Detector

LHCb is a forward-geometry detector with acceptance in the pseudorapidity range $2 < \eta < 5$ [15]. Unlike the general-purpose detectors ATLAS and CMS, LHCb is optimised for the reconstruction of displaced heavy-flavour decays and for precision measurements of hadrons containing b and c quarks.

The detector consists of several subsystems arranged along the beam direction. These include a high-resolution vertex detector close to the interaction point, tracking detectors located on both sides of a dipole magnet, two Ring-Imaging Cherenkov (RICH) detectors used for particle identification, a calorimeter system, and a dedicated muon detector. A schematic illustration of the upgraded configuration used during Run 3 is shown in Fig. 3. The components most relevant to the analysis presented in this work are described below.

Tracking and Momentum Resolution. Precise determination of charged-particle momenta is required in order to resolve the invariant-mass peaks of $\bar{B}^0$ mesons. The tracking system consists of the silicon Vertex Locator (VELO), which surrounds the proton–proton interaction region, an upstream tracking detector, a dipole magnet providing an integrated field of approximately 4 Tm, and tracking stations located downstream of the magnet [16].

This configuration achieves a relative momentum resolution of about 0.4–0.6% [16]. Such resolution leads to sharply reconstructed B meson invariant-mass peaks and significantly improves the ability to distinguish signal candidates from combinatorial background in mass fits.

Vertex Reconstruction. Hadrons containing heavy quarks typically have lifetimes

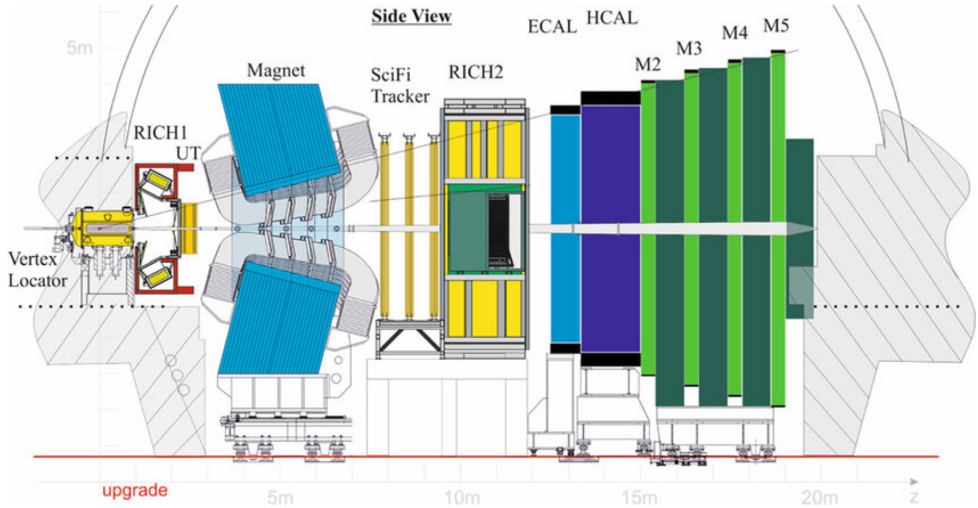

Figure 3: Schematic view of the upgraded LHCb detector operating during Run 3, showing the principal subsystems including the VELO, tracking stations, RICH detectors, calorimeters, and muon system [15].

of a few picoseconds and travel several millimetres before decaying. The VELO measures track impact parameters and decay positions with high spatial precision, enabling the reconstruction of displaced secondary vertices [15]. The resulting vertex resolution is essential for rejecting tracks originating directly from the primary interaction, ensuring that only genuine B-meson decay candidates are kept.

Particle Identification. Efficient separation of pion and kaon particles is essential for this analysis. Charged particles passing through the detector experience a magnetic field from the dipole magnet, causing their trajectories to curve. From this curvature the particle momentum and charge can be determined [16]. However, momentum information alone does not uniquely identify the particle ID.

Two ring-imaging Cherenkov (RICH) detectors provide the information for particle identification (PID) in LHCb [17]. As charged particles propagate through the radiator materials they emit Cherenkov photons, with an emission angle that depends on the particle velocity. By combining the measured Cherenkov angle with the track momentum determined from the tracking system, the mass of the particle can be inferred [18]. This allows efficient separation between π, K, and p over a broad momentum interval.

Calorimeters and Muon System. Energy measurements are provided by a calorimeter system composed of electromagnetic and hadronic sections. These detectors record the energy deposited by particles interacting with the calorimeter material and contribute both to the event reconstruction and to the trigger selection.

At the far end of the detector, a dedicated muon section is installed. It consists of several detection stations separated by absorber layers, allowing muons to penetrate while most hadrons are absorbed [19]. This design enables efficient muon identification with strong suppression of hadronic background. Although the signal decays studied in this analysis are fully hadronic, the muon detector remains essential for reconstructing the normalisation channel $B^+ \to J/\psi(\mu^+\mu^-)K^+$.

B. Run 3 Trigger Architecture

The trigger system determines which proton–proton collision events are recorded for further analysis. During Runs 1 and 2, LHCb used a two-level trigger architecture composed of a hardware stage followed by a software-based high-level trigger [15]. The hardware stage, which operated with an output rate of roughly 1 MHz, relied mainly on information from the calorimeter and muon detectors. In order to satisfy this rate limitation, tight requirements on transverse energy and momentum were applied. Although this configuration was well suited to final states containing leptons, it reduced the acceptance for fully hadronic decay modes and introduced a dependence of the trigger efficiency on the kinematic properties of the reconstructed particles.

The Run 3 upgrade removed the hardware trigger and introduced a triggerless readout system operating at the full bunch-crossing rate of 40 MHz [7]. As a result, detector information from every bunch crossing is read out and the event selection is performed entirely within the software trigger. Events are processed by a two-level software trigger. In the first level (HLT1), a fast reconstruction of tracks and primary vertices is performed, and events containing tracks significantly displaced from the interaction point are selected. The second level (HLT2) reconstructs the event in greater detail, applying enhanced tracking, particle-identification information, and multivariate selection methods [8].

Eliminating hardware thresholds leads to a more uniform trigger efficiency for hadronic decay channels, particularly at lower transverse momentum. For the analysis presented here, an important consequence is that both the fully hadronic signal modes and the muonic normalisation channel $B^+ \to J/\psi(\mu^+\mu^-)K^+$ are selected within the same trigger and reconstruction framework.

C. Data Sample

This analysis uses a subset of proton–proton collision data recorded by the LHCb experiment during the Run 3 data-taking period (2022–2025) at a centre-of-mass energy of $\sqrt{s} = 13.6$ TeV. The dataset corresponds to Blocks 7 and 8 of the centrally produced heavy-flavour selections and includes data collected with both magnet polarities. In Run 3, data-taking is organised into blocks corresponding to periods of stable detector and trigger conditions, ensuring uniform processing and consistent calibrations across the dataset. Combining opposite magnet polarities helps cancel reconstruction asymmetries introduced by the magnetic field configuration. The total integrated luminosity of the sample is approximately 1 fb^{-1}.

The candidates analysed in this work are taken from DecayTree tuples generated after the HLT2 trigger stage [8]. The trigger selects displaced $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ decay topologies, requiring well-reconstructed secondary vertices, good track-fit quality, and significant impact parameters with respect to the primary vertex. Particle-identification information from the RICH detectors is also applied at this stage [18].

After HLT2 selection, events are processed through the Run 3 dataflow model, where they are further reduced and prepared for analysis using the Sprucing framework [20]. The resulting DecayTree tuples contain fully reconstructed heavy-flavour candidates in a compact candidate-based format suitable for physics analysis.

IV. Methodology

The branching-fraction measurement is carried out in two steps. First, signal and normalisation yields are obtained from invariant-mass fits with detailed modelling of backgrounds and cross-feed. Second, these yields are combined with external branching-fraction inputs and relative-efficiency factors to determine $\mathcal{B}(\bar{B}^0 \to D^+\pi^-)$ and $\mathcal{B}(\bar{B}^0 \to D^+K^-)$.

A. Branching-Fraction Extraction

The hadronic branching fractions are measured relative to the decay $B^+ \to J/\psi(\mu^+\mu^-)K^+$. The dimuon final state enables clean reconstruction with well-characterised trigger and tracking efficiencies, and its branching fraction is known with percent-level precision from global averages. The external inputs used are

$$\begin{aligned} \mathcal{B}(B^+ \to J/\psi K^+) &= (1.020 \pm 0.019) \times 10^{-3} \\ \mathcal{B}(J/\psi \to \mu^+\mu^-) &= (5.961 \pm 0.033) \times 10^{-2} \\ \mathcal{B}(D^+ \to K^+\pi^+\pi^-) &= (9.46 \pm 0.24) \times 10^{-2} \end{aligned} \tag{2}$$

taken from the Particle Data Group (PDG) world averages [10].

The hadronic branching fractions are obtained from yield ratios according to Eqs. (3) and (4),

$$\mathcal{B}(\bar{B}^0 \to D^+h^-) = \frac{f_u}{f_d}\,\frac{N_{D^+h^-}}{N_{J/\psi K^+}}\,R_{\mathcal{B}}\,\varepsilon_{\text{rel}}, \tag{3}$$

where $h \in \{\pi, K\}$ and

$$R_{\mathcal{B}} = \frac{\mathcal{B}(B^+ \to J/\psi K^+)\, \mathcal{B}(J/\psi \to \mu^+\mu^-)}{\mathcal{B}(D^+ \to K^+\pi^+\pi^-)}. \tag{4}$$

In Eq. (3), $N_{D^+h^-}$ is the fitted signal yield obtained from the simultaneous $D^+\pi^-/D^+K^-$ mass fit (Sec. IV.E.), and $N_{J/\psi K^+}$ is the fitted normalisation yield obtained from the dedicated $J/\psi K^+$ mass fit (Sec. IV.D.). The factor $\varepsilon_{\rm rel}$ denotes the relative reconstruction efficiency between the signal and normalisation final states. In this work it is treated as a tracking-efficiency correction (Sec. IV.F.), assuming that the efficiencies of the individual charged tracks factorise to a good approximation. As discussed in Sec. II.C., the ratio f_u/f_d enters the branching-fraction determination through Eq. (3). In this analysis it is taken to be unity, $f_u/f_d = 1$, consistent with the approximate equality of the two fragmentation fractions.

B. Overview of the two-fit strategy

The branching-fraction determination in Eq. (3) requires two independent yield measurements. First, a dedicated invariant-mass fit to $B^+ \to J/\psi(\mu^+\mu^-)K^+$ candidates provides the normalisation yield $N_{J/\psi K^+}$. Second, a simultaneous invariant-mass fit to the $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ samples determines the signal yields $N_{D^+\pi^-}$ and $N_{D^+K^-}$. The simultaneous treatment ensures that cross-feed between the two hadronic channels, arising from pion–kaon misidentification, is modelled consistently and constrained using PID calibration information.

The fits are performed separately because the normalisation and signal modes have different final-state topologies and background compositions. The $B^+ \to J/\psi(\mu^+\mu^-)K^+$ decay forms a high-purity dimuon sample with a simple mass model, whereas the fully hadronic $\bar{B}^0 \to D^+h^-$ channels require additional components such as misidentification cross-feed, partially reconstructed backgrounds, and possible Λ_b contributions.

C. Selection observables

Because heavy-flavour hadrons have relatively long lifetimes, their decays frequently occur at secondary vertices displaced from the primary proton–proton interaction point. The compatibility of the decay tracks with a common vertex is assessed through the vertex-fit χ^2/ndf, where smaller values indicate a higher-quality vertex reconstruction. The spatial separation between the reconstructed B candidate and the primary vertex is expressed by the flight-distance significance, FD, while the impact-parameter significance, IP, evaluates how well a track is consistent with its origin from the primary vertex.

The agreement between the reconstructed $\bar{B}^0$ momentum and the vector pointing from the primary to the decay vertex is summarised by DIRA, defined as the cosine of the pointing angle [21]. For true B meson decays this quantity is expected to be close to one.

Additional suppression of combinatorial background is obtained using multivariate classifiers (MVA and BDTG) that combine kinematic and topological variables to distinguish signal from background. These algorithms are widely used within LHCb for displaced b hadron decays [22]. In this analysis the standard LHCb baseline selection is used without modification.

D. Normalisation-channel fit: $B^+ \to J/\psi K^+$

The normalisation yield is obtained from an invariant-mass fit to reconstructed $B^+ \to J/\psi(\mu^+\mu^-)K^+$ candidates. This fit provides the quantity $N_{J/\psi K^+}$ entering Eq. (3).

Candidate reconstruction. Candidates consistent with the decay $J/\psi \to \mu^+\mu^-$ are formed by combining two oppositely charged tracks identified as muons [19]. The invariant mass of the dimuon system is required to be compatible with the nominal J/ψ mass. The B^+ candidate then is constructed by adding a charged kaon track to the J/ψ.

Quality requirements on track fits, vertex reconstruction, and decay topology are applied to ensure a well-defined displaced B^+ decay. The complete set of requirements is summarised in Table 1. These selections suppress background from tracks originating at the primary vertex and reduce combinatorial background arising from random track combinations. The kaon candidate is required to satisfy a kaon-like PID response. This reduces contamination from the Cabibbo-suppressed decay $B^+ \to J/\psi\pi^+$, which would otherwise appear at slightly higher reconstructed mass when the pion is assigned the kaon mass hypothesis.

Table 1: Selection requirements for the $B^+ \to J/\psi K^+$ normalisation channel.

Observable	Requirement
$m_{J/\psi K^+}$	$5100 < m < 5600$ MeV$/c^2$
$m_{\mu^+\mu^-}$	$\lvert m - m_{J/\psi}\rvert < 80$ MeV$/c^2$
Muon ID	muon-ID criteria
Track quality	$\chi^2/\text{ndf} < 10$
B^+ vertex quality	$\chi^2/\text{ndf} < 20$
B^+ flight distance significance	FD > 100
Bachelor PID	PIDK > 5
Track p_T	$p_T > 250$ MeV$/c$

Fit observable. The fit is performed to the invariant mass of the $J/\psi K^+$ in the range 5100–5600 MeV$/c^2$. This mass range is enough to model the smooth combinatorial background while keeping a clean signal region. The signal yields are extracted through an unbinned extended maximum-likelihood fit, where the mass of every candidate is included explicitly in the likelihood.

Signal model. The signal peak in the B^+ invariant-mass distribution is parametrised with a double-sided Crystal Ball function [23]. The Gaussian term represents the detector mass resolution, while the asymmetric tails describe effects such as final-state radiation and residual non-Gaussian resolution. The peak parameter corresponds to the reconstructed B^+ mass.

Background model. The dominant background arises from random combinations of a genuine J/ψ candidate with an unrelated charged track. As this contribution varies smoothly across the fit window, it is modelled with a second-order Chebyshev polynomial. A low-order polynomial is sufficient to describe the smooth combinatorial background in the selected mass region.

Validation. As a cross-check of the signal modelling, a wrong-mass hypothesis study is performed in which the bachelor track is reconstructed with the pion mass. This test is designed to probe possible contamination from $B^+ \to J/\psi\pi^+$ decays.

Fit output. An extended maximum-likelihood fit determines the signal yield $N_{J/\psi K^+}$ together with the background and shape parameters.

E. Simultaneous signal fit: $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$

The signal yields $N_{D^+\pi^-}$ and $N_{D^+K^-}$ are extracted from a single extended maximum-likelihood fit performed simultaneously to the $D^+\pi^-$ and D^+K^- candidate samples. A simultaneous treatment is essential because the two datasets are not independent: residual π^-/K^- misidentification causes a fraction of true $\bar{B}^0 \to D^+\pi^-$ decays to enter the D^+K^- selection and vice versa. A combined likelihood therefore ensures that cross-feed is handled consistently and prevents biases that would arise from fitting the two spectra separately.

Fit observable, range, and binning The fit observable is the reconstructed invariant mass of the D^+h^- system, denoted $m(B)$, evaluated under the mass hypothesis corresponding to the sample definition ($h = \pi$ for $D^+\pi^-$, $h = K$ for D^+K^-).

The fit range $4950 - 6000$ MeV$/c^2$ is chosen to satisfy two conditions. First, it must be wide enough to constrain the smooth combinatorial background from data. Second, it must include the low-mass region populated by partially reconstructed decays in which one or more neutral particles are not reconstructed.

The fit is implemented as a binned extended likelihood using 300 uniform bins across the range. This corresponds to a bin width of approximately 3.5 MeV/c^2, significantly smaller than the detector mass resolution which is typically in the range 12–25 MeV/c^2 for reconstructed B-meson decays [15]. The binning therefore does not distort the signal shape while improving fit stability and computational efficiency for large Run 3 samples.

Baseline selection and sample definition Candidates entering the simultaneous $D^+\pi^-/D^+K^-$ fit are required to satisfy a common set of kinematic, topological and multivariate requirements, summarised in Table 2. The topological selections exploit the long lifetime of the $\bar{B}^0$ meson. Requirements on the vertex quality, flight-distance significance, impact-parameter significance and pointing angle ensure consistency with a displaced decay originating from the primary interaction vertex. These criteria suppress combinatorial background while preserving signal efficiency.

Additional discrimination is provided by two multivariate classifiers. A dedicated $\bar{B}^0$ candidate MVA response and the gradient-boosted decision-tree variable (BDTG). Both are trained to separate signal-like topologies from combinatorial background using kinematic and geometric observables defined in Sec. IV.C. Tight working points are chosen to maximise background rejection while preserving signal efficiency and maintaining identical thresholds in the $D^+\pi^-$ and D^+K^- samples. Applying common multivariate requirements ensures that any difference between the two spectra arises from physics and PID selection rather than from selection biases.

The D^+ meson is reconstructed through $D^+ \to K^+\pi^+\pi^-$. A narrow invariant-mass window around the known D^+ mass, together with particle-identification requirements on the three daughter tracks, enforces consistency with the $K^+\pi^+\pi^-$ hypothesis and suppresses misreconstructed charm backgrounds.

Definition of the $D^+\pi^-$ and D^+K^- samples. After forming the D^+ candidate, the two $\bar{B}^0 \to D^+h^-$ samples are distinguished solely by the particle-identification response of the bachelor track. The discriminant used is PIDK, derived primarily from RICH information [18], which represents the logarithmic likelihood difference between the kaon and pion hypotheses for a given track. A pion-like PIDK requirement defines the $D^+\pi^-$ sample, while a kaon-like requirement defines the D^+K^- sample. The working points are chosen to be mutually exclusive to prevent event overlap, ensuring a statistically well-defined simultaneous fit. The small residual cross-feed arising from π^-/K^- misidentification is not removed but is instead modelled explicitly in the likelihood, allowing it to be constrained by data.

Requirement	Working point
$\bar{B}^0$ mass range	$4950 < m(\bar{B}^0) < 6000$ MeV/c^2
D^+ mass window	$\lvert m(K^-\pi^+\pi^+) - m_{D^+}\rvert < 20$ MeV/c^2
Vertex quality	$\chi^2/\text{ndf}(\bar{B}^0) < 10$, $\chi^2/\text{ndf}(D^+) < 10$
$\bar{B}^0$ flight-distance significance	$\text{FD}(\bar{B}^0) > 70$
$\bar{B}^0$ impact-parameter significance	$\text{IP}(\bar{B}^0) < 15$
$\bar{B}^0$ pointing consistency	$\text{DIRA}(\bar{B}^0) > 0.9995$
Multivariate selection	MVA > 0.9, BDTG > 0.5
D^+-daughter PID (common)	K^-: PIDK > 5; π^+: PIDK < 0 (both pions)
Bachelor PID (sample definition)	$D^+\pi^-$: PIDK < 0; D^+K^-: PIDK > 5

Table 2: Selection requirements for $\bar{B}^0 \to D^+h^-$ candidates used in the simultaneous fit.

Shared and channel-specific signal parameters Both decay modes originate from the same parent particle ($\bar{B}^0$) and are reconstructed in the same detector conditions. The dominant contributions to the peak position and core resolution are therefore common. For this reason, the signal PDFs share a common mean μ and common Gaussian core width σ.

Parameters describing non-Gaussian tails are allowed to differ between the $D^+\pi^-$ and D^+K^- signals. This accounts for the small kinematic differences introduced by the pion and kaon mass assignments, while keeping the mean and resolution parameters common between the two channels.

Signal and combinatorial background modelling The invariant-mass spectra of the signal candidates are parameterised with a double-sided Crystal Ball function [23]. The low-mass tail accounts for final-state radiation and residual energy-losses. The high-mass tail models non-Gaussian resolution effects. This parametrisation describes the asymmetric tails and preserves the clean signal peak shape.

The combinatorial background is modelled with an exponential function, which describes the smooth mass distribution after selection cuts.

Partially reconstructed backgrounds In addition to fully reconstructed decays, the selection also keeps decays in which one or more final-state particles are not reconstructed, usually neutral particles such as photons or π^0 mesons. The missing momentum from these particles is therefore absent from the invariant-mass calculation of the D^+h^- system, resulting in a reconstructed mass below the true $\bar{B}^0$ mass. Partially reconstructed decays therefore populate the low-mass region beneath the signal peak and end at a maximum reconstructed mass below the true B-meson mass. In the absence of detector resolution this boundary would be sharp but finite momentum resolution instead smears the edge into a smooth, asymmetric structure.

These contributions are modelled using analytical endpoint functions, referred to as the HILL and HORNS shapes, implemented as custom classes within the CERN `RooFit` framework [24], with parametrisations following Ref. [25]. The invariant-mass spectrum is determined by the spin and angular distribution of the unreconstructed particle. When the missing particle is a scalar, such as a π^0, the resulting spectrum exhibits a double-peaked structure near the kinematic endpoints, described by the HORNS function. If the missing particle is a vector, such as a photon, the distribution instead forms a single-peaked structure described by the HILL function. Their shape parameters are determined in dedicated single-channel fits and fixed in the simultaneous fit to avoid correlations with the signal yield.

Cross-feed from π^-/K^- misidentification Because the two signal modes differ only by the identity of the bachelor hadron, misidentification of π^- and K^- induces cross-feed between samples. True $D^+\pi^-$ decays can appear in the D^+K^- sample if the pion satisfies the kaon PID requirement, and vice versa. The cross-feed shapes are determined directly from data using a wrong-mass-hypothesis reconstruction. Candidates selected in one sample are re-evaluated by recomputing the bachelor-track four-momentum under the opposite mass assignment ($\pi^- \leftrightarrow K^-$) while keeping the measured three-momentum fixed, and the D^+h^- invariant mass is recalculated. This procedure provides data-driven templates for misidentified decays.

The direction of the peak shift follows from kinematics. Assigning the heavier kaon mass to a true pion increases the reconstructed energy for fixed momentum and shifts the $D^+\pi^- \to D^+K^-$ cross-feed peak above the true D^+K^- signal. Conversely, assigning the pion mass to a true kaon shifts the $D^+K^- \to D^+\pi^-$ cross-feed to lower mass.

PIDCalib inputs The cross-feed yields are not floated freely in the simultaneous fit, as the misidentified components overlap with the signal region and would otherwise be strongly correlated with the true signal yields. Instead, they are constrained relative to the corresponding signal yields through the bachelor-track PID efficiencies. The relations used in the fit are given by Eqs. (5) and (6):

$$N_{\text{mis}}^{D^+\pi^-\to D^+K^-} = N_{\text{sig}}^{D^+\pi^-}\frac{\varepsilon(\pi^-\to K^-)}{\varepsilon(\pi^-\to\pi^-)} \quad (5)$$

$$N_{\text{mis}}^{D^+K^-\to D^+\pi^-} = N_{\text{sig}}^{D^+K^-}\frac{\varepsilon(K^-\to\pi^-)}{\varepsilon(K^-\to K^-)} \quad (6)$$

The fitted quantity $N_{\mathrm{sig}}(D\pi)$ corresponds to the number of true $D\pi$ decays that satisfy the pion-like PID requirement. A fraction of these decays is reconstructed in the DK sample when the bachelor pion is misidentified as a kaon. This fraction is determined by the relative probability of passing the kaon-like PID requirement, which leads directly to the efficiency ratio appearing in Eq. (5). The same reasoning applies to the $D^+K^- \to D^+\pi^-$ cross-feed contribution described by Eq. (6).

The efficiencies entering these relations are obtained from the LHCb Common calibration framework (PIDCalib) [26], which provides data-driven PID measurements from calibration samples in which the particle identity is identified kinematically. The efficiencies are parameterised as functions of track kinematics and reweighted to match the bachelor track kinematics in this analysis.

In the fit, the resulting efficiency ratios are treated as fixed external inputs and are not floated in the likelihood. As a result, the statistical uncertainty returned by the mass fit is determined only by the data.

Λ_b contamination A small peaking contribution from $\Lambda_b^0 \to \Lambda_c^+\pi^-$ is included in the $D^+\pi^-$ spectrum. Although the underlying decay contains a charm baryon rather than a D^+ meson, it can enter the $D^+\pi^-$ sample when the $\Lambda_c^+ \to pK^+\pi^-$ final state is reconstructed under the $D^+ \to K^+\pi^+\pi^-$ hypothesis. If the proton from the Λ_c^+ decay is misidentified as a pion, the three-track combination can satisfy the D^+ mass requirement and daughter PID criteria, forming a candidate compatible with the D^+ topology. Since the bachelor particle in $\Lambda_b^0 \to \Lambda_c^+\pi^-$ is a genuine pion, it also satisfies the pion-like bachelor PID requirement used to define the $D^+\pi^-$ sample, producing a small peaking structure in the reconstructed $m(D^+\pi^-)$ spectrum.

Although the Λ_b^0 mass differs from the $\bar{B}^0$ mass by about 340 MeV, the reconstructed $m(D^+\pi^-)$ distribution can still appear near the $\bar{B}^0$ signal region because the Λ_c^+ candidate is reconstructed under the incorrect D^+ mass hypothesis. Replacing the proton mass with the pion mass changes the reconstructed charm-hadron momentum and shifts the invariant mass of the $D^+\pi^-$ system towards the $\bar{B}^0$ region.

The corresponding contribution to the D^+K^- sample is negligible. The $\Lambda_b^0 \to \Lambda_c^+\pi^-$ decay would require the bachelor pion to be misidentified as a kaon in order to satisfy the kaon-like PID requirement defining the D^+K^- selection, which is strongly suppressed. Although the Cabibbo-suppressed decay $\Lambda_b^0 \to \Lambda_c^+K^-$ could in principle contribute, its branching fraction is much smaller and the expected yield after selection is negligible compared with other backgrounds.

The normalisation of this contribution is obtained from a dedicated control fit in the Λ_b^0 mass window. The control selection follows the same topological and multivariate requirements as the signal selection, but reconstructs Λ_c^+ candidates in a narrow mass window and applies PID requirements appropriate to the $\Lambda_c^+ \to pK^+\pi^-$ final state, including a proton-like requirement on the proton candidate. The Λ_b^0 signal is described with a Johnson S_U PDF [27], while the combinatorial background is modelled with a Bernstein polynomial. Two additional background contributions are included: a partially reconstructed component, dominated by modes such as $\Lambda_b^0 \to \Lambda_c^+\rho^-$ and $\Lambda_b^0 \to \Lambda_c^+\pi^-\pi^0$, described with a HORNS-type shape and a small cross-feed contribution from $\Lambda_b^0 \to \Lambda_c^+K^-$, described by a Gaussian component. The fitted Λ_b^0 signal yield from this control fit defines $N_{\Lambda_b^0}^{\mathrm{ctrl}}$.

The shape entering the $D^+\pi^-$ fit is not taken from simulation but constructed directly from the selected Λ_b^0 control sample. For each accepted Λ_b^0 candidate, the bachelor track is kept with the pion mass hypothesis and the Λ_c^+ candidate is reinterpreted as a D^+ candidate by assigning the pion mass to the proton. The invariant mass $m(D^+\pi^-)$ is then recomputed candidate by candidate, and the resulting distribution is converted into a binned template represented by a `RooHistPdf`. This template therefore provides a data-driven description of the Λ_b^0 contribution under the $\bar{B}^0 \to D^+\pi^-$ hypothesis.

To obtain the contribution entering the $D^+\pi^-$ fit, the control yield is scaled in two steps. First, a re-massing efficiency is computed as the fraction of selected Λ_b^0 candidates whose reconstructed $m(D^+\pi^-)$ falls within the $\bar{B}^0$ fit range, as defined in Eq. (7),

$$\varepsilon_{\mathrm{remass}} = \frac{N_{\bar{B}^0\,\mathrm{win}}}{N_{\Lambda_b^0\,\mathrm{win}}}, \tag{7}$$

where $N_{\bar{B}^0\,\mathrm{win}}$ denotes the number of selected Λ_b^0 candidates entering the $\bar{B}^0$ mass window after reconstruction under the $D^+\pi^-$ hypothesis, and $N_{\Lambda_b^0\,\mathrm{win}}$ the number reconstructed in the Λ_b^0 control window. Second, a PID correction is applied to account for the different daughter-PID requirements in the Λ_c^+ control selection and the misreconstructed D^+ selection. The yield in the $D^+\pi^-$ spectrum is therefore written as Eq. (8),

$$N_{\Lambda_b^0}^{\mathrm{exp}} = N_{\Lambda_b^0}^{\mathrm{ctrl}}\,\varepsilon_{\mathrm{remass}}\,\varepsilon_{\mathrm{PID}}, \tag{8}$$

where $\varepsilon_{\mathrm{PID}}$ denotes the corresponding PID-efficiency ratio, determined from PIDCalib. This contribution is included in the simultaneous fit through a Gaussian constraint.

Fit output The simultaneous extended likelihood fit returns $N_{D^+\pi^-}$ and $N_{D^+K^-}$ together with the background yields and shared peak parameters. These yields are used directly in the branching-fraction calculation presented in Eq. (3).

F. Relative tracking efficiency correction

As introduced in Eq. (3), the branching fraction is determined from the ratio of fitted yields multiplied by a relative efficiency factor, $\varepsilon_{\mathrm{rel}}$. This factor accounts for residual differences in the reconstruction of the charged final states in the signal $\bar{B}^0 \to D^+h^-$ and the muonic normalisation channel. The particle content of the two modes differs. The normalisation decay contains two muons, whereas the signal modes are fully hadronic and include one additional charged track. Because muons and hadrons interact differently with detector material, their tracking efficiencies are not identical and do not fully cancel in the yield ratio.

Single-track efficiencies. Muon tracks define the baseline tracking efficiency, $\varepsilon(\mu^\pm) = 0.969$, obtained from dedicated muon control samples. Charged-hadron efficiencies are written relative to this baseline through a survival factor defined as

$$\varepsilon(h) = \varepsilon(\mu)\, S_h, \qquad S_h = 1 - f_h^{\mathrm{int}}.$$

where f_h^{int} is the hadronic interaction probability and S_h accounts for additional losses from nuclear interactions with detector material [28]. The single-track efficiencies used in the construction of $\varepsilon_{\mathrm{rel}}$ are summarised in Table 3. Expressing the efficiencies in this way allows the common baseline factor $\varepsilon(\mu)$ to cancel in the ratio used to construct $\varepsilon_{\mathrm{rel}}$.

Particle	f_h^{int}	S_h	$\varepsilon(h)$
$\mu^\pm$	—	—	0.969
$\pi^\pm$	0.150	0.850	0.824
$K^\pm$	0.127	0.873	0.846

Table 3: Single-track efficiencies used in the construction of $\varepsilon_{\mathrm{rel}}$.

Construction of $\boldsymbol{\varepsilon_{\mathrm{rel}}}$. The relative tracking factor is defined as

$$\varepsilon_{\mathrm{rel}} = \frac{\varepsilon_{\mathrm{norm}}}{\varepsilon_{\mathrm{sig}}},$$

where each term is expressed as the product of single-track efficiencies for the corresponding final state. Common tracks cancel explicitly in the ratio. In this analysis the tracking efficiencies are written as $\varepsilon_{\mathrm{norm}} = \varepsilon(\mu)^2 \times \varepsilon(K)$ and $\varepsilon_{\mathrm{sig}} = \varepsilon(K) \times \varepsilon(\pi)^2 \times \varepsilon(h)$, where $h \in \{\pi, K\}$. The kaon efficiency cancels between numerator and denominator. Substituting $\varepsilon(h) = \varepsilon(\mu)\, S_h$ for hadrons, all common factors are removed except for one power of $\varepsilon(\mu)$. This residual term arises because the signal final state contains one additional charged track relative to the normalisation topology (four tracks versus three), so the cancellation is not exact. The relative tracking efficiency therefore becomes

$$\varepsilon_{\rm rel}(D^+\pi^-) = \frac{1}{S_\pi^3\,\varepsilon(\mu)} \tag{9}$$

$$\varepsilon_{\rm rel}(D^+K^-) = \frac{1}{S_\pi^2\,S_K\,\varepsilon(\mu)} \tag{10}$$

The resulting values of $\varepsilon_{\rm rel}$, obtained using the single-track inputs discussed above, are presented in the next section.

V. Results

This section presents the results of the two-stage fitting procedure described in the previous chapter. The outcomes of the $B^+ \to J/\psi K^+$ normalisation fit and the simultaneous fit to the $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ samples are shown, from which the signal yields are extracted.

A. Normalisation-Channel Fit Results: $B^+ \to J/\psi K^+$

The invariant-mass distribution of selected $B^+ \to J/\psi K^+$ candidates, after the requirements described in Sec. IV.D. and summarised in Table 1, is shown in Fig. 4. A clear and isolated signal peak is observed near the known B^+ mass.

The fit is performed using a Double Crystal Ball function to model the signal component and a second-order Chebyshev polynomial to describe the combinatorial background. The fitted signal yield and key shape parameters are summarised in Table 4.

Table 4: Fitted parameters from the $B^+ \to J/\psi K^+$ normalisation fit. Uncertainties are statistical only.

Parameter	Value
$N_{\rm sig}$	136071 ± 428
$N_{\rm bkg}$	288551 ± 686
μ [MeV]	5281.49 ± 0.06
σ_L [MeV]	10.17 ± 0.06
σ_R [MeV]	3.00 ± 0.21
α_L	5.000 ± 0.008
n_L	2.07 ± 0.04
α_R	0.73 ± 0.01
n_R	50.0 ± 1.9

The fitted mass peak is $\mu = 5281.49 \pm 0.06$ MeV, which is compatible with the Particle Data Group value $m_{B^+}^{\rm PDG} = 5279.34 \pm 0.12$ MeV [10]. The observed $\sim$2 MeV shift is consistent with the known LHCb momentum-scale uncertainty. A relative momentum bias of a few 10^{-4} propagates approximately linearly to the reconstructed mass, implying shifts of order $m_B \times 10^{-4} \sim$ 1–2 MeV for fully reconstructed B decays [29].

The fitted Gaussian core width, $\sigma_L = 10.17 \pm 0.06$ MeV, is consistent with the expected $\mathcal{O}(10\,\text{MeV})$ invariant-mass resolution for $B^+ \to J/\psi K^+$ decays at LHCb. The asymmetric parameter $\sigma_R = 3.00 \pm 0.21$ MeV indicates a sharper Gaussian fall-off on the high-mass side of the peak. Non-Gaussian behaviour is described by the Crystal Ball tail parameters (α_L, n_L) and (α_R, n_R). The low-mass tail is primarily associated with QED final-state radiation in $J/\psi \to \mu^+\mu^-$ decays, which shifts reconstructed candidates towards

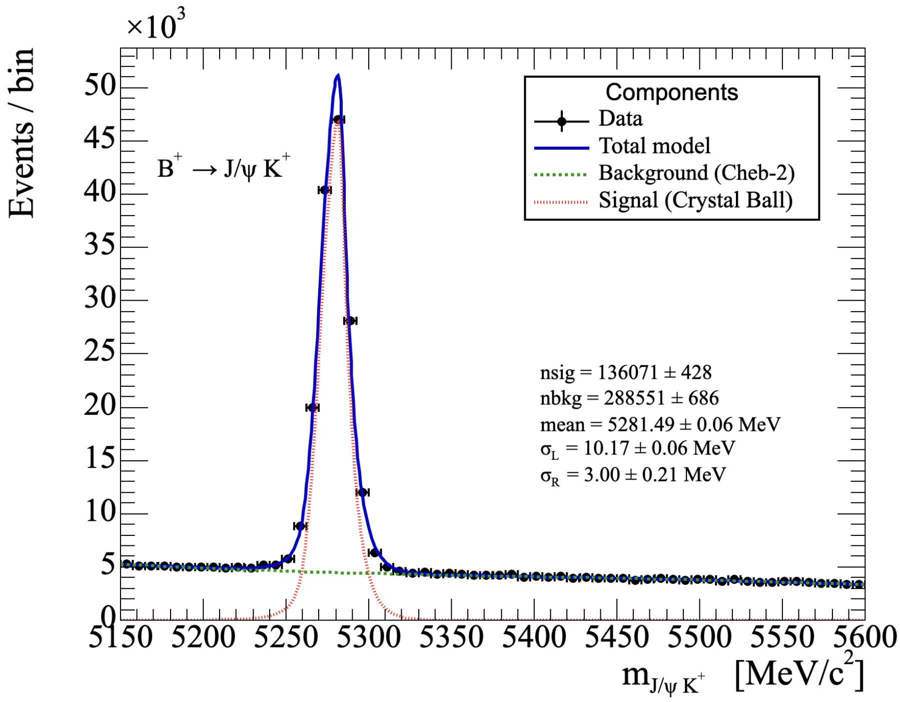

Figure 4: Invariant-mass distribution of $B^+ \to J/\psi K^+$ candidates with fit overlay. The signal component is shown in red, the background component in green, and the total model in blue.

lower masses [30]. The fitted tail parameters are consistent with expectations, where the low-mass tail arises primarily from final-state radiation while the high-mass side remains close to Gaussian.

Within the chosen signal window, the signal-to-background ratio is high, and the statistical precision on the normalisation yield is better than 0.4%. The fit model describes the data well across the full mass range.

As an additional validation, a wrong-mass hypothesis test was performed by reconstructing candidates under the pion mass assumption for the bachelor track. No significant peaking structure is observed within the kaon signal region, confirming that misidentified reflections do not contaminate the normalisation yield.

B. Simultaneous Fit Results

The simultaneous fit described in Sec. IV.E. is applied to the selected $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ samples. The resulting invariant-mass fits are shown in Fig. 5. Both linear and logarithmic representations are provided to illustrate the behaviour in the signal region and in the low mass region.

The simultaneous likelihood fit converges successfully with fit status 0 and covariance quality 3, indicating reliable parameter uncertainties. The estimated distance to minimum (EDM) is below 2×10^{-4}, confirming stable convergence. The shared mass parameters are determined to be $\mu = 5280.238 \pm 0.024$ MeV$/c^2$ and $\sigma =$

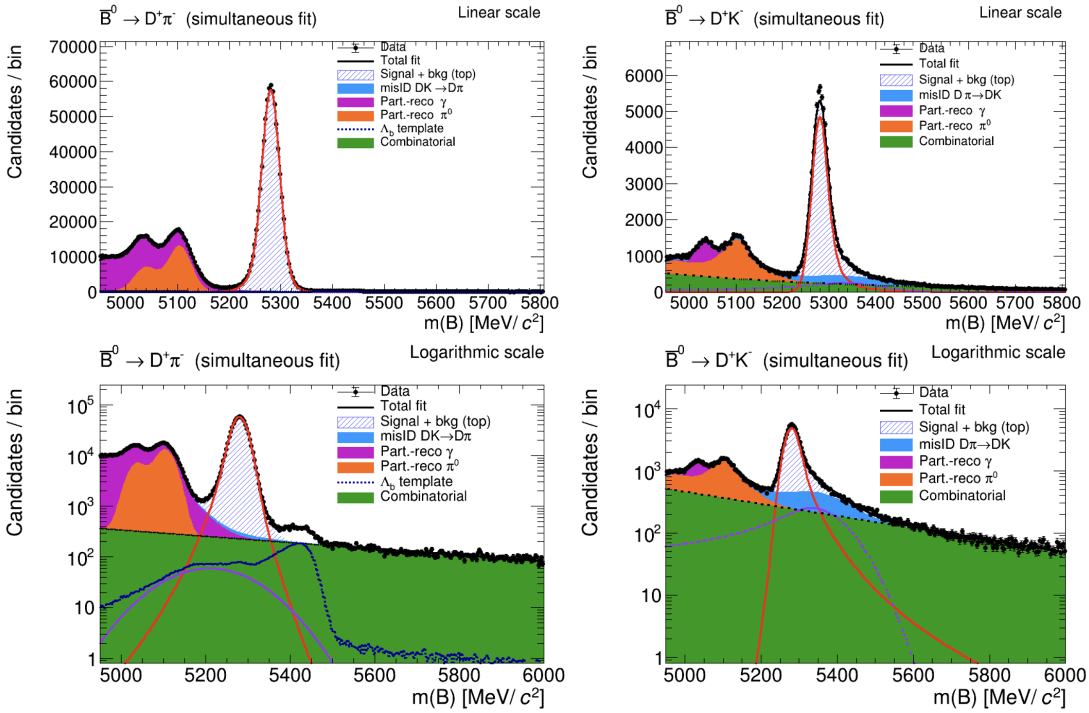

Figure 5: Simultaneous invariant-mass fits to the $\bar{B}^0 \to D^+\pi^-$ (left) and $\bar{B}^0 \to D^+K^-$ (right) samples. The top row shows the fits in linear scale, while the bottom row shows the same distributions in logarithmic scale. Individual components are displayed separately.

17.502 ± 0.021 MeV$/c^2$. The fitted peak position is consistent with the world-average $\bar{B}^0$ mass, $m_{\bar{B}^0}^{\rm PDG} = 5279.65 \pm 0.12$ MeV$/c^2$, as reported by the Particle Data Group [10]. The measured mass resolution of σ = 17.5 MeV$/c^2$ is consistent with expectations for fully reconstructed hadronic B decays in LHCb. Since the invariant mass is calculated from charged-track four-momenta, its resolution is dominated by the tracking momentum resolution, which is approximately 0.4–0.6% [16]. A naive scaling, $\delta m_B \sim m_B(\delta p/p)$, for $m_B \approx 5280$ MeV$/c^2$, suggests a resolution of order 20 MeV$/c^2$. The fitted value therefore lies well within the expected range. These results indicate that the chosen model provides a good description of the data.

Extracted Yields The fitted component yields are summarised in Table 5. A clear hierarchy is observed between the Cabibbo-favoured and Cabibbo-suppressed modes (Sec. II.A.):

$$N(D^+\pi^-) \gg N(D^+K^-).$$

To an approximation, the ratio of partial widths scales as

$$\frac{\Gamma(\bar{B}^0 \to D^+K^-)}{\Gamma(\bar{B}^0 \to D^+\pi^-)} \sim \left|\frac{V_{us}}{V_{ud}}\right|^2 \left(\frac{f_K}{f_\pi}\right)^2,$$

where the decay-constant factor captures the dominant SU(3)-breaking in the colour-allowed tree amplitude (up to small form-

Component	$D^+\pi^-$ yield	D^+K^- yield
Signal	$752\,673 \pm 933$	$69\,045 \pm 439$
Mis-ID cross-feed	$4\,228 \pm 27$	$20\,536 \pm 26$
Part-reco (γ-like)	$415\,298 \pm 1\,117$	$6\,704 \pm 176$
Part-reco (π^0-like)	$315\,795 \pm 905$	$37\,997 \pm 500$
Combinatorial	$56\,985 \pm 1\,095$	$59\,395 \pm 803$
Λ_b^0 contribution	$10\,774 \pm 347$	–

Table 5: Component yields obtained from the simultaneous fit.

factor and phase-space differences) [3]. Using the PDG values for V_{us} and V_{ud} together with the lattice average f_K/f_π [31] this estimate gives an expected ratio of $\mathcal{O}(0.08)$, which is compatible with the observed ratio of fitted yields

$$\frac{N(D^+K^-)}{N(D^+\pi^-)} = 0.0917 \pm 0.0006_{\text{stat}}$$

once differences in reconstruction and particle-identification efficiencies between the two channels are taken into account.

The direction of the misidentification peak shifts is purely kinematic. The reconstructed invariant mass depends on the assumed mass of the bachelor track through $E = \sqrt{p^2 + m^2}$. Since $m_K > m_\pi$, assigning the kaon mass hypothesis to a true pion increases the reconstructed energy and shifts the mass to higher values, so the $D^+\pi^- \to D^+K^-$ reflection appears to the right of the D^+K^- peak. Conversely, assigning the pion mass to a true kaon reduces the reconstructed energy, shifting the $D^+K^- \to D^+\pi^-$ reflection to lower mass, to the left of the $D^+\pi^-$ peak.

The relative sizes of the two cross-feed components are dependent on the applied PID working points. In the selected kinematic region, the pion-like requirement (PIDK < 0) admits a larger fraction of true kaons than the kaon-like requirement (PIDK > 5) admits true pions. This asymmetry is reflected quantitatively in the fitted yields, which imply an effective misID rate of approximately 6% for $K^- \to \pi^-$ compared to about 3% for $\pi^- \to K^-$. Consequently, the $DK \to D\pi$ cross-feed contribution exceeds the $D^+\pi^- \to D^+K^-$ component despite the smaller underlying D^+K^- signal yield, in agreement with the PIDCalib efficiency inputs used in the fit.

Partially reconstructed and combinatorial backgrounds. Both channels exhibit partially reconstructed backgrounds below the signal peak. The γ-like components populate the region just beneath the signal with a smooth hill shape, characteristic of decays with a single missing photon, while the π^0-like components produce a broader double-peaked structure extending across the low-mass region, as expected for scalar missing particles described by the

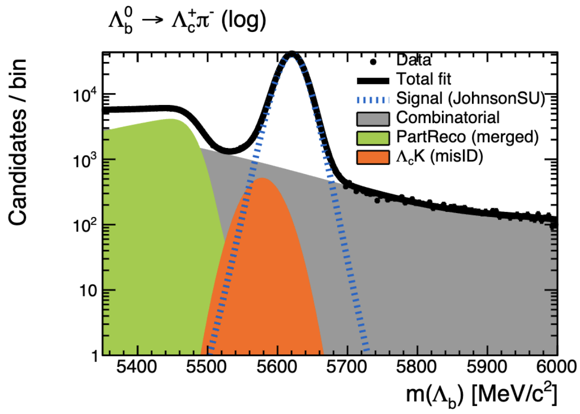

Figure 6: Control fit to the $\Lambda_b^0 \to \Lambda_c^+\pi^-$ mass spectrum. The logarithmic projection is shown for clarity, highlighting the lower-mass background structure. The fitted Johnson S_U signal component and background contributions are overlaid.

HORNS shape.

The overall composition of the two samples is similar, although the $D^+\pi^-$ spectrum contains a somewhat larger partially reconstructed component, consistent with the Cabibbo-favoured $b \to c\bar{u}d$ transition. In the D^+K^- channel these contributions are reduced, reflecting the suppression of the corresponding $b \to c\bar{u}s$ processes. The combinatorial background in both channels is well described by a smooth exponential shape with no residual peaking structure over the fitted mass range.

Λ_b component validation. The control fit to the $\Lambda_b^0 \to \Lambda_c^+\pi^-$ mass spectrum is shown in Fig. 6. The signal peak is narrow and well isolated, with the Johnson S_U component providing an accurate description of its asymmetric core and non-Gaussian tails. The logarithmic projection demonstrates that the model also reproduces the lower-mass region well, where contributions from partially reconstructed decays and the small $\Lambda_c^+K^-$ cross-feed become visible, while the combinatorial component remains smooth across the fitted range.

The fitted shape parameters extracted from this control sample are subsequently used to construct the template describing Λ_b^0 contamination in the $\bar{B}^0 \to D^+\pi^-$ fit. Under the $\bar{B}^0 \to D^+\pi^-$ mass hypothesis, this background maps to a broad structure in the intermediate-mass region of the $D^+\pi^-$ spectrum. The corresponding enhancement is visible in the logarithmic projection of Fig. 5 and is well reproduced by the Λ_b^0 template.

Although the overall yield of this contribution is small compared to the $D^+\pi^-$ signal, explicitly modelling it prevents distortions of the combinatorial background and improves the stability of the simultaneous fit. Its inclusion therefore ensures that potential Λ_b^0 contamination is accounted for in a controlled manner within the global likelihood model.

C. Determination of the branching fractions

Using the values in Table 3, inserted in Eqs. (9) and (10), the resulting corrections are

$$\varepsilon_{\rm rel}(D^+\pi^-) = 1.680 \pm 0.025_{\rm exp}$$
$$\varepsilon_{\rm rel}(D^+K^-) = 1.636 \pm 0.020_{\rm exp}$$

Here "exp" denotes the experimental uncertainty. Values greater than unity are expected, reflecting the lower overall tracking efficiency of the fully hadronic signal relative to the muonic normalisation channel. The uncertainty on $\varepsilon_{\rm rel}$ arises from the material-interaction fractions reported in Ref. [28]. These are propagated as an experimental uncertainty on the branching fractions.

The fitted signal yields (Tables 4 and 5) and the external branching fractions entering the normalisation (Eq. 2) are substituted into Eq. 4 to obtain the ratio of branching fractions. These results are then inserted into Eq. 3 to extract the absolute branching fractions.

Luminosity correction. The normalisation fit was performed on a subset of data, corresponding to an integrated luminosity of 403.468 pb^{-1}, whereas the $D^+\pi^-/D^+K^-$ simultaneous fit uses the Block7 and Block8 samples, with a combined luminosity of 1112.6 pb^{-1} as obtained from the LHCb Run Database [32]. To ensure consistency, the normalisation yield is scaled by the relative luminosity factor

$$f_{\mathcal{L}} = \frac{1112.6}{403.468} = 2.76,$$

before evaluating Eqs. 4 and 3. This procedure assumes that the trigger, reconstruction, and selection efficiencies are sufficiently stable across the data-taking blocks and magnet polarities used in the analysis, so that the signal yields scale approximately linearly with the integrated luminosity. Since the same trigger and offline selections are used for both datasets, large efficiency variations are not expected. Within the scope of this project, any residual differences are assumed to be negligible compared to the dominant statistical and external uncertainties and are therefore not included in the luminosity scaling.

Final results. After propagating statistical uncertainties from the fitted yields, experimental uncertainties from detector and efficiency corrections, and external uncertainties from the input branching fractions, the measured values are

$$\mathcal{B}_{D^+\pi^-} = 2.165 \pm 0.007 \pm 0.032 \pm 0.069$$
$$\mathcal{B}_{D^+K^-} = 1.933 \pm 0.014 \pm 0.024 \pm 0.062$$

expressed in units of 10^{-3} for $\mathcal{B}_{D^+\pi^-}$ and 10^{-4} for $\mathcal{B}_{D^+K^-}$. Here $\mathcal{B}_{D^+\pi^-} \equiv \mathcal{B}(\bar{B}^0 \to D^+\pi^-)$ and $\mathcal{B}_{D^+K^-} \equiv \mathcal{B}(\bar{B}^0 \to D^+K^-)$. The quoted uncertainties correspond, in order, to statistical, experimental-efficiency, and propagated external-input contributions. Additional modelling effects related to assumptions are discussed in Sec. VI. but are not included as separate numerical uncertainties.

Comparison with world averages. The current world averages reported by the Particle Data Group [10] are

$$\mathcal{B}(\bar{B}^0 \to D^+\pi^-)_{\text{PDG}} = (2.51 \pm 0.08) \times 10^{-3},$$
$$\mathcal{B}(\bar{B}^0 \to D^+K^-)_{\text{PDG}} = (2.05 \pm 0.08) \times 10^{-4}.$$

The relative differences are found to be

$$\Delta_{\%}(D^+\pi^-) = -13.76 \pm 4.11\ \%,$$
$$\Delta_{\%}(D^+K^-) = -5.69 \pm 4.94\ \%.$$

The D^+K^- result is consistent with the PDG world average within uncertainties. For the $D^+\pi^-$ mode, the central value lies below the PDG average, with a difference at the level of about 3σ when the quoted uncertainties are combined. This may indicate the impact of residual modelling effects not included as separate systematic uncertainties in the present analysis.

In addition to the individual branching fractions, the ratio $\frac{\mathcal{B}_{D^+K^-}}{\mathcal{B}_{D^+\pi^-}}$ provides a complementary comparison, since several systematic effects and external inputs partially cancel in the ratio. Using the central values obtained in this work, the ratio is determined to be

$$\frac{\mathcal{B}_{D^+K^-}}{\mathcal{B}_{D^+\pi^-}} = 0.0893 \pm 0.0007 \pm 0.0017 \pm 0.0040,$$

where the uncertainties are statistical, experimental, and external, respectively.

The PDG world average for this ratio is $(8.19 \pm 0.20) \times 10^{-2}$ [10], while the most precise single measurement to date is the LHCb result $(8.22 \pm 0.11_{\text{stat}} \pm 0.25_{\text{syst}}) \times 10^{-2}$. The statistical uncertainty achieved here, 7×10^{-4}, is approximately a factor of two smaller than that of the previous measurement, illustrating the statistical precision achievable with the Run 3 dataset.

The value of the ratio obtained in this work is slightly higher than the current world average. When the total uncertainties are combined, the difference corresponds to a deviation of approximately $1.4\,\sigma$ with respect to the PDG value. This shift is primarily driven by the smaller central value obtained for the $\bar{B}^0 \to D^+\pi^-$ branching fraction relative to the PDG average, which increases the ratio $\mathcal{B}_{D^+K^-}/\mathcal{B}_{D^+\pi^-}$.

VI. Systematic Uncertainties and Modelling

The quoted branching-fraction results include statistical uncertainties and propagated uncertainties from tracking-efficiency corrections and external branching-fraction inputs. These uncertainties are grouped into statistical, experimental, and external. Additional modelling-related systematic effects are discussed qualitatively below but are not included as separate numerical contributions in the quoted uncertainties, as their evaluation would require dedicated systematic studies beyond the scope of the present analysis.

A. Modelling Assumptions

The D^+K^- final state shares a kaon with the normalisation channel, $J/\psi K^+$, leading to partial cancellation of kaon-related tracking and PID effects in the branching-fraction ratio. In contrast, the $D^+\pi^-$ channel depends more strongly on pion-specific efficiencies, for which such cancellations are weaker. This means that the $D^+\pi^-$ branching fraction is particularly sensitive to remaining efficiency mismatches between the two decay channels, which arise from differences in the kinematics of their final-state particles. This is one possible source of the larger fractional shift observed in $D^+\pi^-$, but a quantitative assessment would require dedicated reweighting and systematic studies.

Efficiency corrections are applied as functions of momentum and pseudorapidity and are assumed to be valid for both bachelor and daughter tracks. However, these tracks originate from different decay vertices and therefore populate different regions of phase space. Imperfect modelling of these spectra can modify the effective efficiency correction, which can be tested by reweighting simulated kinematic spectra to match those observed in data.

B. PID Calibration and Efficiency Modelling

Particle-identification (PID) efficiencies enter the simultaneous fit through constraints on cross-feed backgrounds derived from PIDCalib samples. Simulation mismatches between calibration and signal kinematics, binning choices, or wrong-mass template assumptions could affect the constrained cross-feed yields.

In addition, $D^+\pi^-$ simulation is used as an approximation for D^+K^- kinematics in the absence of dedicated Run 3 D^+K^- Monte Carlo. Since PID efficiencies depend strongly on particle momentum, differences between pion and kaon spectra may introduce additional modelling effects. These are not assigned separate numerical uncertainties in the present study but would be evaluated in a full systematic analysis.

C. Fit-Model Considerations

The invariant-mass fits rely on specific parameterisations for signal and background components. Variations of background shapes, signal tail parameters, and cross-feed constraints were explored to assess the stability of the fitted yields. Refitting the data with alternative polynomial and Chebychev combinatorial background models resulted in changes to the fitted signal yields of less than 0.3% for $D^+\pi^-$ and less than 2% for D^+K^-. The larger variation in the D^+K^- channel is expected, as the smaller sample is more sensitive to background modelling. These variations are treated as stability checks rather than a full estimate of the fit-model systematic uncertainty, and no additional numerical uncertainty is assigned.

The Λ_b^0 mass-window mapping assumes that the efficiency derived from the control-sample remassing procedure is applicable to the signal selection. A quantitative evaluation of this assumption would require dedicated control studies and is therefore not included as a separate uncertainty.

D. Tracking and External Inputs

Tracking-efficiency corrections enter through the relative efficiency factor $\varepsilon_{\rm rel}$, derived from LHCb material-interaction studies. The associated uncertainties are propagated to the branching fractions and contribute to the experimental uncertainty.

External branching fractions are treated as fixed inputs, and their quoted uncertainties are propagated directly. These external inputs constitute the dominant non-statistical contribution to the total uncertainty.

VII. Interpretation and Outlook

This section discusses the projected impact of the full Run 3 dataset and places the present measurement within the broader context of existing experimental results and theoretical expectations.

A. Projected statistical precision at 30 fb^{-1}

The simultaneous $D^+\pi^-/D^+K^-$ analysis presented in this work uses the Block 7+8 data-taking periods, corresponding to an integrated luminosity of approximately 1 fb^{-1}. To estimate the statistical sensitivity achievable with the full Run 3 dataset, results are projected to a total of 30 fb^{-1}.

The evolution of the delivered luminosity during Run 3 is shown in Fig. 7, as reported by the LHCb online operations monitoring. The figure illustrates the cumulative integrated luminosity delivered in recent years at $\sqrt{s} = 13.6$ TeV. In particular, it demonstrates that Run 3 has already reached integrated luminosities well above 10 fb^{-1} per year, with a cumulative total approaching the projected 30 fb^{-1} scale. The ~ 1 fb^{-1} analysed here therefore represents only an early subset of the available Run 3 dataset. This motivates a projection to the full dataset expected by the end of the run.

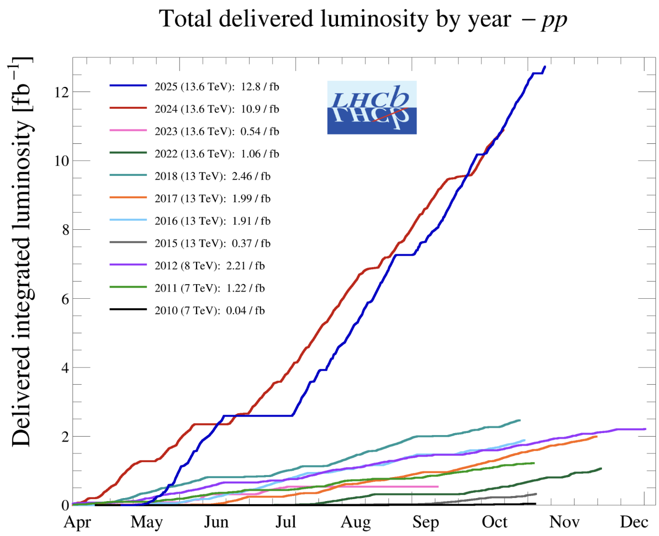

Figure 7: Cumulative delivered luminosity in pp collisions as reported by the LHCb operations monitoring [33]. This motivates the projection of statistical sensitivity from the analysed ~ 1 fb^{-1} (Block 7+8) dataset to 30 fb^{-1}.

Assuming the analysis strategy and selection performance remain unchanged, the statistical component of the branching-fraction uncertainty is expected to scale approximately as

$$\sigma_{\text{stat}}(\mathcal{B}) \propto \frac{1}{\sqrt{\mathcal{L}}}$$

where $\mathcal{L}$ denotes the integrated luminosity of the Run 3 data. Relative to a 1 fb^{-1} baseline, an increase to 30 fb^{-1} implies an improvement by a factor $\sqrt{30} \simeq 5.48$ in the statistical uncertainty.

Using this scaling, the projected statistical uncertainties at 30 fb^{-1} are

$$\sigma_{\text{stat}}\left[\mathcal{B}(\bar{B}^0 \to D^+\pi^-)\right] = 0.0013 \times 10^{-3},$$
$$\sigma_{\text{stat}}\left[\mathcal{B}(\bar{B}^0 \to D^+K^-)\right] = 0.0025 \times 10^{-4}.$$

To quantify the impact relative to existing measurements, the PDG [10] lists the most precise single determination of $\mathcal{B}(\bar{B}^0 \to D^+\pi^-)$ with a statistical uncertainty of 0.01×10^{-3}. Comparing to the projected 30 fb^{-1} precision from this study, the statistical uncertainty is reduced by a factor

$$\frac{0.01}{0.0013} \approx 7.7,$$

corresponding to an improvement in relative statistical precision from $\sim 0.40\%$ to $\sim 0.06\%$. A similar improvement is expected for the D^+K^- mode. This level of statistical precision would significantly improve the sensitivity of both decay channels in precision flavour studies.

B. Context within $b \to c\bar{u}q$ Measurements

Figure 8 compares the present determination of the branching fractions for $\bar{B}^0 \to D^+\pi^-$ and $\bar{B}^0 \to D^+K^-$ with previous measurements and Standard Model (SM) expectations. The main panel displays the current PDG averages together with selected experimental results from Belle (2022) and LHCb (2013), as well as SM/QCDF predictions for each decay mode. The purple markers correspond to the values obtained in this work using 1 fb^{-1} of data, where the quoted uncertainties represent the total uncertainty. The red markers indicate the projected precision at 30 fb^{-1}, assuming that the non-statistical contributions remain unchanged. In this panel, the red error bars denote the total projected uncertainty, while the black overlay represents the statistical component alone.

Two qualitative features are evident. First, the experimental measurements shown lie below the central SM/QCDF expectations for both decay modes, with the separation more pronounced in the $D^+\pi^-$ channel. Second, even at 30 fb^{-1} the total projected uncertainty is dominated by the external contribution, such that increasing the integrated luminosity alone does not substantially reduce the overall precision unless the external inputs are improved.

The measurement presented in this work provides an independent Run 3 determination using the upgraded LHCb detector and the fully software-based trigger, allowing previous results to be tested in a new experimental environment. The projected Run 3 dataset also significantly reduces the statistical uncertainty, as illustrated by the reduced uncertainty bands in Fig. 8.

C. Future Work

Projection to 30 fb^{-1} indicates that the overall precision of this measurement will remain dominated by external inputs rather than by statistical uncertainty alone. Improved control of these inputs will therefore be required to fully exploit the increased Run 3 statistics. Several improvements could further strengthen the analysis, including more precise efficiency corrections using larger PID calibration samples, improved modelling of partially reconstructed backgrounds, and dedicated simulated samples for the D^+K^- channel.

The study could also be extended to decay modes such as $\bar{B}^0 \to D^{*+}h^-$ or combined with flavour-tagging to test CP asymmetries and other observables. More generally, further dedicated studies could investigate the origin of the $b \to c\bar{u}q$ anomaly by examining possible experimental systematic effects, refining theoretical predictions within the QCDF framework, and exploring potential contributions from physics beyond the Standard Model.

VIII. Conclusions

A simultaneous determination of $\mathcal{B}(\bar{B}^0 \to D^+\pi^-)$ and $\mathcal{B}(\bar{B}^0 \to D^+K^-)$ has been performed using 1 fb^{-1} of LHCb Run 3 data,

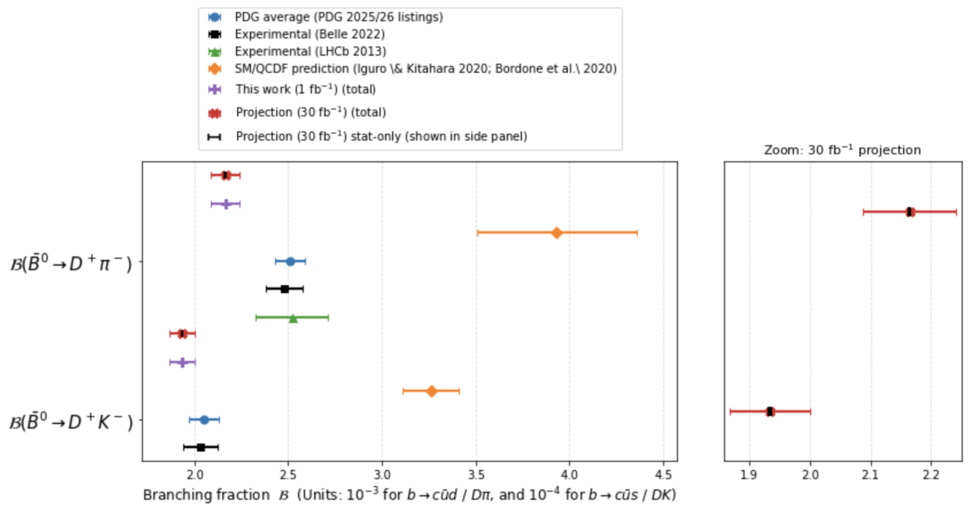

Figure 8: Comparison of experimental measurements and SM/QCDF expectations for $\mathcal{B}(\bar{B}^0 \to D^+\pi^-)$ and $\mathcal{B}(\bar{B}^0 \to D^+K^-)$. The right-hand side panel provides a zoom of the projection region and overlays the statistical component only (black), illustrating the relative size of the statistical and total uncertainties.

normalised to $B^+ \to J/\psi K^+$. The branching fractions obtained within the modelling and efficiency assumptions described in this work are

$$\mathcal{B}_{D^+\pi^-} = 2.165 \pm 0.007 \pm 0.032 \pm 0.069,$$
$$\mathcal{B}_{D^+K^-} = 1.933 \pm 0.0134 \pm 0.024 \pm 0.063,$$

expressed in units of 10^{-3} and 10^{-4}, respectively. The quoted uncertainties correspond to statistical, experimental efficiency, and propagated external-input contributions. The D^+K^- result is compatible with the current world average within uncertainties, while the $D^+\pi^-$ result lies noticeably below the PDG average. The total uncertainty is dominated by external inputs associated with the normalisation branching fractions rather than by the statistical precision of the dataset.

In the context of the $b \to c\bar{u}q$ tension, this measurement provides an independent Run 3 benchmark obtained using a consistent experimental framework. Projections to the full 30 fb^{-1} Run 3 dataset indicate a statistical-uncertainty improvement approaching an order of magnitude relative to the most precise existing single measurements. These results establish a framework for future high-precision studies of tree-level $b \to c\bar{u}q$ dynamics using the complete Run 3 dataset.

References

[1] T. Gershon et al. "Testing the Standard Model with CP asymmetries in flavour-specific non-leptonic decays". In: (2021). arXiv: `2111.04478 [hep-ph]`.

[2] M. Beneke and M. Neubert. "QCD factorization for $B \to PP$ and $B \to$

PV decays". In: *Nucl. Phys. B* 675 (2003), p. 333. DOI: 10.1016/j.nuclphysb.2003.09.026.

[3] M. Beneke and M. Neubert. "QCD factorization for $B \to D\pi$ and $B \to DK$ decays". In: *Nucl. Phys. B* 651 (2003), p. 225. DOI: 10.1016/S0550-3213(02)01059-2.

[4] S. Iguro and T. Kitahara. "Implications for new physics from a novel puzzle in $\bar{B}^0_{(s)} \to D^+_{(s)}\{\pi^-, K^-\}$ decays". In: *Phys. Rev. D* 102 (2020), p. 071701.

[5] M. Bordone et al. "A puzzle in $\bar{B}^0_{(s)} \to D^+_{(s)}\{\pi^-, K^-\}$ decays and extraction of f_s/f_d". In: *Eur. Phys. J. C* 80 (2020), p. 951.

[6] R. Aaij et al. "Measurement of the ratio of branching fractions $\mathcal{B}(B^0 \to D^+K^-)/\mathcal{B}(B^0 \to D^+\pi^-)$". In: *Phys. Rev. D* 87 (2013), p. 092010.

[7] LHCb Collaboration. *LHCb Trigger and Online Upgrade Technical Design Report*. Tech. rep. CERN-LHCC-2014-016. CERN, 2014.

[8] R. Aaij et al. "The LHCb trigger and its performance in Run 3". In: *JINST* 19 (2024), P02020.

[9] L. Scarabotto. "The LHCb Upgrade I Trigger". In: *Proceedings of the 41st International Conference on High Energy Physics (ICHEP 2022)*. Vol. ICHEP2022. PoS. 2022, p. 395. DOI: 10.22323/1.414.0395. arXiv: 2212.11061 [hep-ex].

[10] Particle Data Group. "Review of Particle Physics". In: *Prog. Theor. Exp. Phys.* (2024), p. 083C01.

[11] M. Bauer, B. Stech, and M. Wirbel. "Exclusive nonleptonic decays of D, D_s and B mesons". In: *Z. Phys. C* 34 (1987), p. 103. DOI: 10.1007/BF01561122.

[12] A. V. Manohar and M. B. Wise. *Heavy Quark Physics*. Cambridge University Press, 2000.

[13] R. Aaij et al. "Study of the kinematic dependence of f_s/f_d in pp collisions". In: *Phys. Rev. D* 87 (2013), p. 092010.

[14] L. Evans and P. Bryant. "LHC Machine". In: *JINST* 3 (2008), S08001.

[15] R. Aaij et al. "LHCb detector performance". In: *Int. J. Mod. Phys. A* 30 (2015), p. 1530022. DOI: 10.1142/S0217751X15300227.

[16] R. Aaij et al. "Performance of the LHCb tracking system". In: *JINST* 9 (2014), P09007. DOI: 10.1088/1748-0221/9/09/P09007.

[17] R. Aaij et al. "Performance of the LHCb RICH detector". In: *Eur. Phys. J. C* 73 (2013), p. 2431. DOI: 10.1140/epjc/s10052-013-2431-9.

[18] R. Aaij et al. “Performance of the LHCb RICH detectors during Run 2”. In: *JINST* 18 (2023), P05026. DOI: 10.1088/1748-0221/18/05/P05026.

[19] R. Aaij et al. “Performance of the muon identification at LHCb”. In: *JINST* 10 (2015), P02007. DOI: 10.1088/1748-0221/10/02/P02007.

[20] A. Abdelmotteleb, A. Bertolin, C. Burr, et al. “The LHCb Sprucing and Analysis Productions”. In: *Computing and Software for Big Science* 9 (2025), p. 15. DOI: 10.1007/s41781-025-00144-5.

[21] CMS Collaboration. “CMS Physics Technical Design Report, Volume II: Physics Performance”. In: *Journal of Physics G: Nuclear and Particle Physics* 34.6 (2007), pp. 995–1579. DOI: 10.1088/0954-3899/34/6/S01.

[22] V. V. Gligorov and M. Williams. “Efficient, reliable and fast high-level triggering using a boosted decision tree”. In: *JINST* 8 (2013), P02013. DOI: 10.1088/1748-0221/8/02/P02013.

[23] T. Skwarnicki. *A study of the radiative cascade transitions between the* Υ' *and* Υ *resonances*. Tech. rep. DESY-F31-86-02. DESY, 1986.

[24] W. Verkerke and D. P. Kirkby. *The RooFit toolkit for data modeling*. 2003. arXiv: physics/0306116.

[25] R. Aaij et al. “Measurement of CP observables in $B^\pm \to D^{(*)}K^\pm$ and $B^\pm \to D^{(*)}\pi^\pm$ decays”. In: *Phys. Lett. B* 777 (2018), pp. 16–30. DOI: 10.1016/j.physletb.2017.11.070. arXiv: 1708.06370 [hep-ex].

[26] L. Anderlini et al. *The PIDCalib package*. Tech. rep. LHCb-PUB-2016-021. CERN, 2016.

[27] N. L. Johnson. “Systems of frequency curves generated by methods of translation”. In: *Biometrika* 36 (1949), pp. 149–176.

[28] LHCb Collaboration. *Material interaction and hadronic absorption study for tracking efficiency*. accessed 12 February 2026. URL: https://twiki.cern.ch/twiki/bin/viewauth/LHCb/TrackingEffAbsLength.

[29] R. Aaij et al. “Precision measurement of the Λ_b^0, B^0 and B_s^0 hadron masses”. In: *Phys. Rev. Lett.* 110 (2013), p. 182001.

[30] E. Barberio and Z. Was. “PHOTOS: A universal Monte Carlo for QED radiative corrections in decays”. In: *Comput. Phys. Commun.* 79 (1994), p. 291.

[31] S. Aoki et al. “FLAG Review 2019”. In: *Eur. Phys. J. C* 80 (2020), p. 113. DOI: 10.1140/epjc/s10052-019-7354-7.

[32] LHCb Collaboration. *LHCb Run Database (LBRUNDB)*. accessed 15 February 2026. URL: https://lbrundb.cern.ch/.

[33] LHCb Collaboration. *LHCb Online Operations Plots – Delivered Luminosity*. accessed 18 February 2026. URL: https://lbgroups.cern.ch/online/OperationsPlots/index.htm.